Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Friction

MARKSCHEME ($g = 9.8\,\mathrm{m\,s^{-2}}$)

---

1. Limiting $F = \mu R = 19.6$; applied $10 < 19.6 \Rightarrow F = 10$ **M1A1** [3]
2. $F = \mu mg = 8.82$ **M1A1** [3]
3. Limiting 9.8; net 5.2; $a = 1.3$ **M1A1** [4]
4. Compare $mg\sin\alpha$ with $\mu mg\cos\alpha$; slips **M1A1** [4]
5. $a = g(\sin 30^\circ - 0.4\cos 30^\circ)$ **M1A1** [5]
6. $F = mg\sin\alpha - \mu mg\cos\alpha$ (or 0 if already holds) **M1A1** [5]
7. Limiting 11.76; net 8.24; a **M1A1** [4]
8. $\mu = \frac{12}{mg}$ **M1A1** [3]
9. $mg\sin\alpha = \mu mg\cos\alpha \Rightarrow \tan\alpha = \mu$ **M1A1** [3]
10. Retardation $g(\sin\alpha + \mu\cos\alpha)$; $v^2 = u^2 + 2as$ **M1A1** [6]
11. Resolve: $R = mg - P\sin\theta$; $P\cos\theta - F = ma$ **M1A1** [5]
12. $F = \min(mg\sin\alpha, \mu mg\cos\alpha)$ **M1A1** [4]
13. $P\cos\theta = \mu(mg - P\sin\theta)$; solve P **M1A1** [5]
14. $a = g(\sin\alpha - \mu\cos\alpha) = 2 \Rightarrow \mu = \ldots$ **M1A1** [5]
15. Coulomb: $F \le \mu R$; independent of area; opposes motion **B1** [3]
16. Up: retardation $g(\sin\alpha + \mu\cos\alpha)$; down: $g(\sin\alpha - \mu\cos\alpha)$; energy/SUVAT **M1A1** [7]
17. $P\cos\alpha = \mu(mg - P\sin\alpha)$ **M1A1** [5]
18. Net force up plane $= P - mg\sin\alpha - \mu mg\cos\alpha$; a **M1A1** [5]
19. Moments + resolve; compare F needed with μR **M1A1** [6]
20. $a = -\mu g$; $0 = u^2 + 2as \Rightarrow s = \frac{u^2}{2\mu g}$ **M1A1** [4]
21. $a = g(\sin\alpha - \frac{1}{3}\tan\alpha\cos\alpha) = g(\sin\alpha - \frac{1}{3}\sin\alpha) = \frac{2}{3}g\sin\alpha$ **M1A1** [4]
22. $a = \frac{P - mg\sin\alpha - \mu mg\cos\alpha}{m}$ **M1A1** [5]
23. $P = \mu mg$; $Q\cos 30^\circ = \mu(mg - Q\sin 30^\circ)$; eliminate μ **M1A1** [6]
24. $mg\sin\alpha + \mu mg\cos\alpha = 5$? Or F down $= 5$; solve μ **M1A1** [5]
25. Up: $a = g(\sin 45^\circ + \mu\cos 45^\circ) = g\sqrt{2}$; down $g(\sin 45^\circ - \mu\cos 45^\circ) = \frac{g}{\sqrt{2}}$ **M1A1** [6]